

1. Objectives:

1. To understand how 1D Array works in Java.
2. To understand how 2D Array works in Java.
3. To learn how to do sum, average, minimum, and maximum using arrays.

2. Introduction:

An array is a collection of elements of the same data type stored in contiguous memory locations. In Java, arrays can be one-dimensional (1D) or multi-dimensional (2D), depending on the requirement. Loop structures such as for loops are commonly used to access and manipulate array elements efficiently. A one-dimensional array is used to store a list of values, while a two-dimensional array is used to represent data in tabular form, such as rows and columns. By using loops and arrays together, programmers can perform various operations like input, output, calculation of sum, average, and finding minimum and maximum values.

3. Required Hardware and Software:

- **Hardware:** AMD RYZEN 7, RAM 8 GB
- **Software:** Antigravity

4. Experiment Implementation:

Experiment 1: 1D Array

```
package ForthLab;

import java.util.Scanner;

public class ClassOne {

    public static void main(String[] args) {

        int[] a = new int[10];
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter numbers: ");

        for (int i = 0; i < 10; i++) {
            a[i] = sc.nextInt();
        }
        for (int i = 0; i < 10; i++) {
            System.out.print(a[i] + " ");
        }
        sc.close();
    }
}
```

3.1: Experiment 1 output

```
PS D:\BAUST CSE\Java Programing\4th_Lab> & 'C:\Program Files\Java\jdk-25.0.2\bin\java.
'-cp' 'C:\Users\Rezwan\AppData\Roaming\Antigravity\User\workspaceStorage\aab67e00475b;
7fc2187\bin' 'ForthLab.ClassOne'
Enter numbers: 1 2 3 4 5 6 7 8 9 0
1 2 3 4 5 6 7 8 9 0
PS D:\BAUST CSE\Java Programing\4th_Lab> █
```

Experiment 2: Sum, Average, Minimum and Maximum

```
package ForthLab;

import java.util.Scanner;

public class ClassTwo {

    public static void main(String[] args) {
```

```

int sum = 0;
int[] a = new int[10];
Scanner sc = new Scanner(System.in);

System.out.print("Enter numbers: ");

for (int i = 0; i < 10; i++) {
    a[i] = sc.nextInt();
}
for (int i = 0; i < 10; i++) {
    sum = sum + a[i];
}
System.out.println("Sum = " + sum);

float avg = sum / 10.0f;
System.out.println("Average = " + avg);

int min = 99999, max = 0;
for (int i = 0; i < 10; i++) {
    if (min > a[i]) {
        min = a[i];
    }
}
System.out.println("Min = " + min);

for (int i = 0; i < 10; i++) {
    if (a[i] > max) {
        max = a[i];
    }
}
System.out.println("Max = " + max);

sc.close();
}
}

```

3.2: Experiment 2 output

```

PS D:\BAUST CSE\Java Programing\4th_Lab> & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' '-XX:+ShowCodeDetailsIn
'-cp' 'C:\Users\Rezwan\AppData\Roaming\Antigravity\User\workspaceStorage\aab67e000475b74918c49dc1f105108a\redhat.ja
• 7fc2187\bin' 'ForthLab.ClassTwo'
Enter numbers: 1 2 3 4 5 6 7 8 9 0
Sum = 45
Average = 4.5
Min = 0
Max = 9
PS D:\BAUST CSE\Java Programing\4th_Lab> 

```

Experiment 3: 2D Array

```

package ForthLab;

import java.util.Scanner;

public class ClassThree {

    public static void main(String[] args) {
        System.out.print("Enter 9 numbers: ");

        Scanner sc = new Scanner(System.in);

        int[][] a = new int[3][3];

        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                a[i][j] = sc.nextInt();
            }
        }

        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                System.out.print(a[i][j] + "\t");
            }
        }
    }
}

```

```
        System.out.println();
    }
    sc.close();
}
}
```

3.3: Experiment 3 output

```
● PS D:\BAUST CSE\Java Programing\4th_Lab> & 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe' '-XX:+S
'-cp' 'C:\Users\Rezwan\AppData\Roaming\Antigravity\User\workspaceStorage\aab67e00475b74918c49dc1f1
7fc2187\bin' 'ForthLab.ClassThree'
Enter 9 numbers: 1 2 3 4 5 6 7 8 9 0
1      2      3
4      5      6
7      8      9
○ PS D:\BAUST CSE\Java Programing\4th_Lab> 
```

5. Discussion & Conclusion:

In this lab, we learned arrays in Java through practical implementation. One-dimensional arrays were used to store multiple values and perform operations such as displaying elements, calculating sum and average, and finding minimum and maximum values. Additionally, two-dimensional arrays were used to represent and display data in matrix form using nested loops. Through these programs, the use of loops, array indexing, and logical operations became clear. Overall, this lab helped in developing a strong understanding of array manipulation and improved problem-solving skills in Java programming.

7. Reference:

1. <https://www.geeksforgeeks.org/java/arrays-in-java/> accessed at 6.45 PM on 14th April 2026.
2. <https://www.geeksforgeeks.org/java/multidimensional-arrays-in-java/> accessed at 7.25 PM on 14th April 2026.